

KLU-6o Lube Oil Unit Operation & Maintenance

Platform Alpha — document control cover (session insert)

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This cover page is a session insert. Following pages are excerpted from the source publication listed in SOURCES.md.

KLS-KLU60-OM — Lube oil unit operating targets (session facts)

Applies to tag **50-UZ-105** (KLU-60). Oil grade: **ISO VG 46**.

Parameter	Target / limit
Cleanliness (ISO 4406)	17/15/12
Water — alarm	200 ppm
Water — action	500 ppm (investigate cooler integrity)

Cooler tube leak is a common cause of water ingress. Excess water elevates vibration and bearing temperature in connected machines (50-PA-101A / 50-PA-101B).

RECLAMATION

Managing Water in the West

Facilities, Instructions, Standards, and Techniques
Volume 2-4

Lubrication of Powerplant Equipment



U.S. Department of the Interior
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LUBRICATION OF POWERPLANT EQUIPMENT

purification schedule. By keeping track of the condition of the oil, damage to equipment caused by oil deterioration can be prevented. There are a variety of tests that can be performed on an oil depending on its type and service. Some tests can be performed in the field to obtain a quick indication of an oil's condition, but since field testing is not as complete or as accurate as laboratory analysis, laboratory tests should be performed as well.

5.1.1.1 Sample Collection. For the oil analysis to be effective, the sample must be representative of the oil in the system. A sample skimmed off the top of an oil tub after the oil has cooled and contaminants have settled out, may test cleaner than the oil actually is. Conversely, a sample taken from a drain line at the bottom of an oil tub will likely contain sediment and contaminants that have built up over time, providing a much worse picture of the oil condition. Ideally, the oil sample should be drawn from the middle of the oil tub while the unit is operating. In most cases, this is not practical, and other procedures are required. The procedure used will vary depending on whether the system is circulating.

Circulating systems have a continuous flow of oil provided by a pump. An example of this in a hydroplant is a turbine guide bearing where the oil is supplied through the side of the bearing and flows out the top and bottom of the bearing to return to the sump. In a circulating system, it is best to obtain a sample from the supply pipe to the bearing. The sampling port should be from an area where there is turbulence in the pipe, such as at or directly after an elbow or other fitting. This will ensure that the oil is well mixed. The sampling port should be downstream from the pump but before any inline filters.

The most common noncirculating systems in hydroelectric plants are the generator guide and thrust bearing. Gear boxes, as found on gate operators, are another example. The bearings or gears are submerged or partially submerged in a tub of oil with no external pump to circulate the oil. Also, thrust bearings typically have hydrostatic lubrication through the high-pressure lubrication system, but these are operated only during startup and shutdown. When the unit is operating, oil in the tub circulates because of the rotating shaft. Once the unit is shut down, any contaminants are going to separate out of the oil. To get a representative sample of the oil, the sample should be taken while the unit is operating or shortly after shutdown. The most common place to take a sample on a noncirculating system is from the drain. Even with the unit running, a sample drawn from the drain will probably get a higher concentration of contaminants than the oil lubricating the bearing. If this is the only place to obtain a sample, it is best to wait until the unit is shut down and the bearing is being drained for filtering. The bearing should be drained while the oil is still warm. To flush the line and sediment that may be near the drain, at least 1/4 of the oil should be drained out of the tub before the sample is taken.

LUBRICATION OF POWERPLANT EQUIPMENT

If the sampling is to be accomplished on a regular basis between filtering, a permanently mounted tube extending into the bearing housing should be used. The tubing can be installed on the bottom or the side of the oil tub and extend into the oil near the bearing that is being lubricated. A sample drawn from this area of the tub will be more representative of the oil actually lubricating the bearing. There are vacuum devices available that can assist in obtaining the sample.

When using a permanently mounted sample tube, it is important to flush the tube before taking the sample. At least five times the dead volume of the tube should be flushed before the sample is taken. This flushing should remove any contaminants that may have settled in the tube.

The bottle used to take the sample must be very clean to prevent contaminating the sample. ISO 3722 is the standard for the cleanliness of sample bottles. For most applications in hydroplants, a bottle meeting the ISO 3722 requirements for "Super Clean" is recommended. There are several options for bottle material. The type of material is not important. Clear materials, such as glass and PET plastic, allow a visual inspection of the oil.

There are numerous sizes of bottles available. The size required depends on the tests that are going to be performed. The size of the bottle should be coordinated with the testing laboratory based on the volume of oil that they require.

5.1.1.2 Field Tests. A visual inspection of an oil sample is the simplest type of field test. The sample to be inspected should be stored at room temperature away from direct sunlight for at least 24 hours before the inspection. The sample should then be checked for sediment, separated water, unusual color or cloudiness, and any unusual odors. For comparison, it is a good idea to keep a sample of new, unused oil of the same type and manufacturer stored in a sealed container in a cool dark place. The used sample can then be compared to the new sample with respect to color, odor, and general appearance.

Hazy or cloudy oil may be the result of water contamination. The crackle test can be used to verify the presence of water in oil, but does not give any quantitative results. The crackle test can be conducted by making a small cup from aluminum foil, adding a few drops of the oil, and heating rapidly with a small flame. The test can also be conducted by immersing a hot soldering iron in a sample of the oil. In either method, an audible crackling sound will be heard if water is present. Eye protection should be worn during the test because oil may splatter while being heated.

LUBRICATION OF POWERPLANT EQUIPMENT

7.2 Turbine Oil

Under normal conditions, the lubricating oil for a hydroelectric unit's guide and thrust bearings experiences relatively mild service, but it is expected to have a long service life. To have a long life, a high quality oil with various additives to enhance and maintain its quality is required.

In most powerplants, a highly refined turbine oil is used for bearing lubrication. Table 3 lists some typical properties of a Group I turbine oil. A Group II oil would have better oxidation stability. The oil should be rust and oxidation inhibited with an antifoam additive. The oil should also be resistant to emulsification and separate readily from water. Antiwear or extreme pressure additives are not required or desired.

Table 3.—Typical Turbine Oil Specifications

TYPICAL SPECIFICATIONS FOR TURBINE OIL				
Characteristics	ISO VISCOSITY GRADE			ASTM TEST NO.
	32	46	68	
Viscosity, centistokes @ 40 °C	28.8 – 35.2	41.4 –50.6	61.2 – 74.8	D 445
Viscosity Index, minimum	94	94	94	D 2270
Pour Point, °C maximum	-6	-6	-6	D 97
Flash Point, °C minimum	180	180	180	D 92
Total Acid Number, mg KOH/g	Report	Report	Report	D 974
Rust Preventive Characteristics	Pass	Pass	Pass	D 665A
Foaming Characteristics Sequence 1, mL maximum	50/0	50/0	50/0	D 892
Air Release, 50 °C, minutes max	5	7	10	D 3427
Emulsion Characteristics @ 54 °C, minutes to 3 mL emulsion max	30	30	30	D 1401
Oxidation Stability Hours to neut. No. 2.0 min Minutes to 175 kPa drop, min	2000 350	2000 350	2000 175	D 943 D 2272

The recommended oil viscosity is usually specified by the equipment manufacturer and depends on the operating speed, load, and temperature as well as the bearing clearances. The most common viscosities used in turbines are the ISO viscosity grades 32, 46, and 68.

7.3 Hydraulic Systems

The primary purpose of hydraulic fluid is to transmit power. To accomplish this effectively, the fluid must be incompressible and flow readily through the system. The fluid must also have sufficient viscosity to seal and lubricate the components of the hydraulic system. There are a variety of fluids capable of performing these functions, but the most satisfactory hydraulic fluid is usually oil.